

Case study 1:-

Retail Billing System :- A supermarket wants to automate customer billing. Develop a program to accept prices of three products, calculate the subtotal, GST (18%), and generate the final bill.

Algorithm

1. Start.
2. Declare variables: milk, bread, rice, subtotal, gst, finalBill.
3. Input the prices of Milk, Bread, and Rice.
4. Calculate **Subtotal**:
 - $\text{subtotal} = \text{milk} + \text{bread} + \text{rice}$
5. Calculate **GST (18%)**:
 - $\text{gst} = \text{subtotal} \times 18 / 100$
6. Calculate **Final Bill**:
 - $\text{finalBill} = \text{subtotal} + \text{gst}$
7. Display Subtotal, GST, and Final Bill.
8. Stop.

Pseudocode:-

BEGIN

DECLARE milk, bread, rice AS FLOAT

DECLARE subtotal, gst, finalBill AS FLOAT

INPUT milk

INPUT bread

INPUT rice

subtotal = milk + bread + rice

gst = subtotal * 18 / 100

finalBill = subtotal + gst

PRINT "Subtotal = ", subtotal

PRINT "GST = ", gst

PRINT "Final Bill = ", finalBill

END

Test cases:- input/output combination.

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_1.c -o CASE_study_1 ; if ($?) { .\CASE_study_1 }
enter the price of milk 55
enter the price of bread 35
enter the price of rice 350
subtotal=440final_gst=79.200005final_bill=519.200012
PS C:\Users\khush\Desktop\PROJECT 1>
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_1.c -o CASE_study_1 ; if ($?) { .\CASE_study_1 }
enter the price of milk 70
enter the price of bread 50
enter the price of rice 300
subtotal=420final_gst=75.600006final_bill=495.600006
PS C:\Users\khush\Desktop\PROJECT 1>
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_1.c -o CASE_study_1 ; if ($?) { .\CASE_study_1 }
enter the price of milk 60
enter the price of bread 45
enter the price of rice 250
subtotal=355final_gst=63.900002final_bill=418.899994
PS C:\Users\khush\Desktop\PROJECT 1>
```